Matt Frisbie

Building Browser Extensions

Create Modern Extensions for Chrome, Safari, Firefox, and Edge

Foreword by Stefan Aleksic and Louis Vilgo, cofounders of Plasmo

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To my parents Mona and Patrick and my sister Lauren.You are all wonderful. I will continue to fill your bookshelves until you tell me to stop.

To Jordan. Your unwavering support advances the world of web development and keeps the corners of our carpets nice and flat.

Foreword

We are Stefan and Louis, co-founders of Plasmo. We first started our company building browser extensions for cyber security, experiencing firsthand the many difficulties extension developers face. We quickly realized that the process of building an extension was way too difficult. We decided to open source our framework, and dedicated our company's vision to solving the toughest problems in browser extension development. As of today, our framework is the most popular browser extension SDK worldwide, and our fast-growing extension product suite is used by some of the world's leading extension developers.

A large community soon gathered and shared even more issues blocking their development with us. We were excited to hear that Matt was writing a book on the subject! Extension development is tricky and filled with poor documentation, difficulty getting started, a massive transition from one extension version to another looming, and much more. There has never been a comprehensive guide to extension development, but Matt's book is changing that.

People use browsers for most of their lives. You might use your browser to read the news or find a good breakfast burrito recipe when you wake up. Software engineers might use their browsers to review pull requests and test their front end. Sales representatives use their browsers to send outbound messages to prospective clients on LinkedIn, and security engineers use their browsers to review new phishing alerts. These use cases are different, but each person uses the same tool to do their job.

Browser extensions augment the browser and transform it from a generic tool into a highly specialized one. A sales representative has different needs compared to a security engineer. In a few years, sales representatives and security engineers will have a stack of browser extensions that automate, categorize, and track their work. Some professions have picked up on this faster than others. Ask a high-performing sales representative about browser extensions they have installed, and you might hear a list of more than ten.

Browser extensions are potent but underutilized tools. The lack of know-how and a complete manual such as Matt's book has curbed people's imagination of what their browser could be. As knowledge of building extensions increases, there will be a tremendous appetite for new browser extensions that serve all kinds of niches we have never seen before.

This book encapsulates years of experience and research. It digs deep into critical details you could only find in random forums, obscure documentation, and fragmented source code online. It is everything you need to know about building browser extensions, from a well-rounded beginner-friendly introduction to a comprehensive guide for veterans. We especially love the following chapters:

- – Chapter **4**, "Browser Extension Architecture," is for adept developers who have yet to work with browser extensions.
- – Chapter **6**, "Understanding the Implications of Manifest V3," is for developers who need immediate help transitioning their extension to the newer manifest version.
- – Chapter **16**, "Tooling and Frameworks," is a must-read for developers who want to supercharge their workflow with robust frameworks and DevOps/CI automation to produce enterprise-grade browser extensions.

We can't wait to use your extension one day!

Stefan and Louis

Co-founders of Plasmo

Introduction

The world of browser extensions has far more than meets the eye. Consider the following:

- On average, 40% of Internet users in the United States use an ad-blocker on any device; overwhelmingly, these adblockers take the form of browser extensions.
- The tech company Honey, whose primary product is a browser extension, was acquired by PayPal in 2020 for $4 billion.
- As of 2021, there were 1.8 million apps in Apple's App Store; the Chrome Web Store has 180,000 extensions.

When I saw there were 0 relevant Amazon search results for "build chrome extension," I nearly fell out of my chair. I knew at once that this book *must* be written.

Building Browser Extensions: Create Modern Extensions for Chrome, Safari, Firefox, and Edge covers all the knowledge you will need to write cross-browser extensions with the latest web development tools. Browser extensions are given access to *extremely* powerful APIs. I believe most developers are blind to that power – and unaware of just how much it is within their reach.

This book is designed to enlighten web developers and illuminate the true potential of the browser extension software platform. It is geared for developers who have experience building websites and can apply their knowledge to a new software domain. This book is not ideal for people new to programming – it would be like an inexperienced cook starting off by learning to make a sauce.

A major barrier to developing browser extensions is the appalling status quo of documentation. The fragmentation between different browsers and different manifest versions turns slogging through the documentation into a mind-numbingly onerous affair. I wrote this book specifically to address this problem. The reader will learn what is possible with the APIs, how they can best be applied, and all the traps to avoid. The book is not intended to *replace* the API documentation, as it is changing all the time. Instead, it is intended to supplement the API documentation; the book has plenty of direct links to the Chrome Developers and MDN sites throughout.

The transition to manifest v3 is upon us, and already it is causing problems. If you are confused about what manifest v3 is, what are its implications, and how best to navigate the ongoing transition, this book is for you. I dedicated an entire chapter to the manifest v2/v3 transition.

The lingua franca of web development is React, and this book gives special attention to the best ways in which you can write a browser extension in React. It also covers all the supplemental tools you'll need along the way, such as Webpack, Parcel, and Plasmo.

Like many developers, I learn by example. I was annoyed that so many APIs listed in the documentation were totally inscrutable. For example, the omnibox API is amazing and incredibly useful, but the documentation on how to use it is *garbage*. I just wanted a simple example to pick apart and play with, and there was nothing to be found. To fill this need, I created a companion extension for the book: *Browser Extension Explorer*. It's an open source browser extension with dozens of interactive demos. Each demo shows how various browser extension pieces and APIs work, and each includes links to the specific source files so you can see how it was built.

You can download *Browser Extension Explorer* from the Chrome Web Store, or find a link to it on the companion website to this book: `https://buildingbrowserextensions.com`.

Any source code or other supplementary material referenced by the author in this book is available to readers on GitHub via the book's product page, located at https://github.com/Apress/Building-Browser-Extensions-by-Matt-Frisbie. For more detailed information, please visit http://www.apress.com/source-code.

Acknowledgments

I'd like to acknowledge the tremendous work done by editors Divya Modi, Shonmirin P.A., James Markham, and everyone else involved with the book's production. A capable and agile staff does so much to improve the quality of the product. You were all invaluable resources, and it was a pleasure working with you.

I'd also like to thank the book's technical reviewer Jeff Friesen. Technical authors are nothing without this meticulous work behind the scenes, and his contributions were nothing short of outstanding.

Finally, I would like to thank Apress for publishing this book with me. This was my first title where the book was entirely my own concept, and I am thankful that it found a home with such a terrific publisher.

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Matt Frisbie

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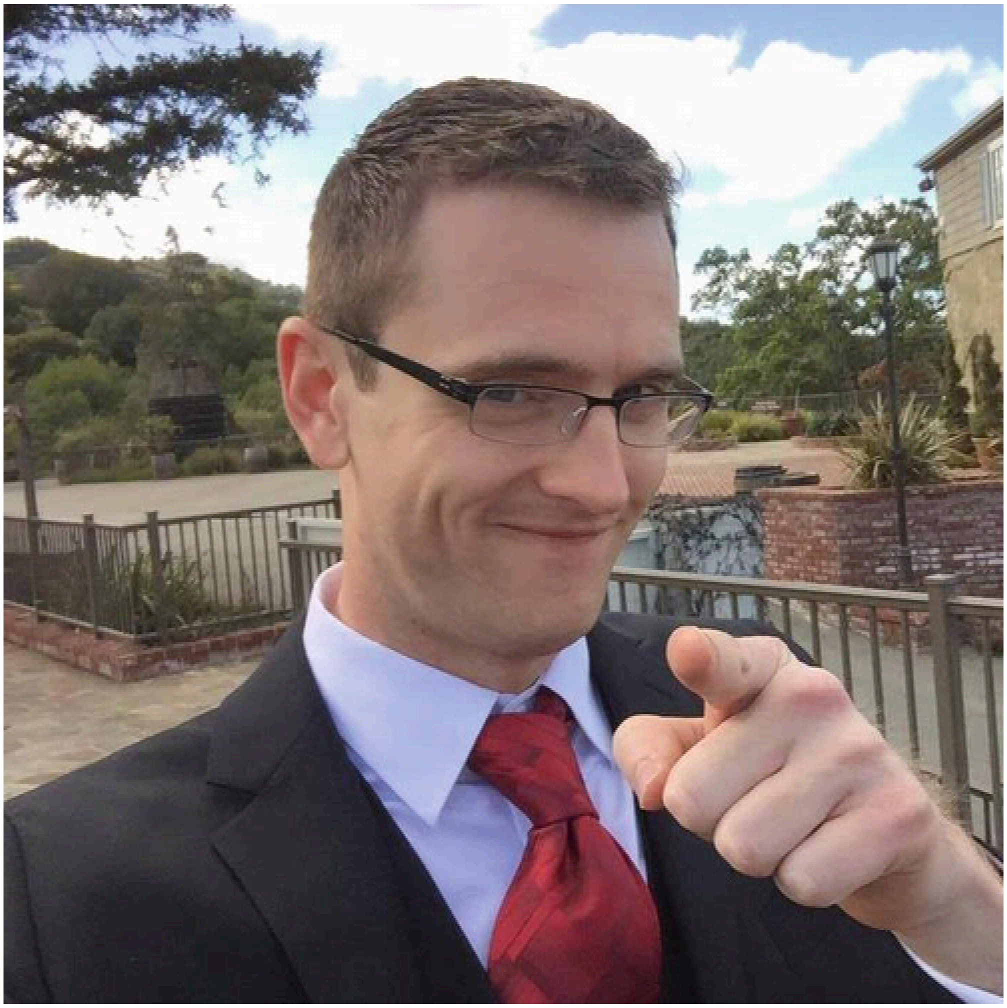

worked in web development for over a decade. During that time, he's been a startup co-founder, an engineer at a Big Four tech company, and the first engineer at a Y Combinator startup that would eventually become a billion-dollar company. As a Google software engineer, Matt worked on both the AdSense and Accelerated Mobile Pages (AMP) platforms; his code contributions run on most of the planet's web browsing devices. Prior to this, Matt was the first engineer at DoorDash, where he helped lay the foundation for a company that has become the leader in online food delivery. Matt has written three books, *Professional JavaScript for Web Developers*, *Angular 2 Cookbook*, and *AngularJS Web Application Development Cookbook*, and recorded two video series, "Introduction to Modern Client-Side Programming" and "Learning AngularJS." He speaks at frontend meetups and webcasts, and is a level 1 sommelier. He majored in Computer Engineering at the University of Illinois Urbana-Champaign. Matt's Twitter handle is @mattfriz.

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